Lesson 4 — Classification and Evaluation Metrics

Technologies for Artificial Intelligence

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Before we start

- Exercise 3 was due today
- Ridge, Lasso, and the choice of the penalty strength

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Today: from a number to a decision

- Logistic regression: the model and why the cost had to change
- The confusion matrix, and why accuracy lies
- Precision, recall, F1 — and the threshold nobody chose
- Receiver operating characteristic (ROC) curves, and where they mislead
- Choosing a threshold from money

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The question has changed

- Lesson 3: given a house, **how many euros?**
- Lesson 4: given a disk drive, **will it fail?**

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The fleet

- **8,000 disk drives** in a data centre
- Six SMART counters each — Self-Monitoring, Analysis and Reporting Technology
- Label: did it fail within thirty days?

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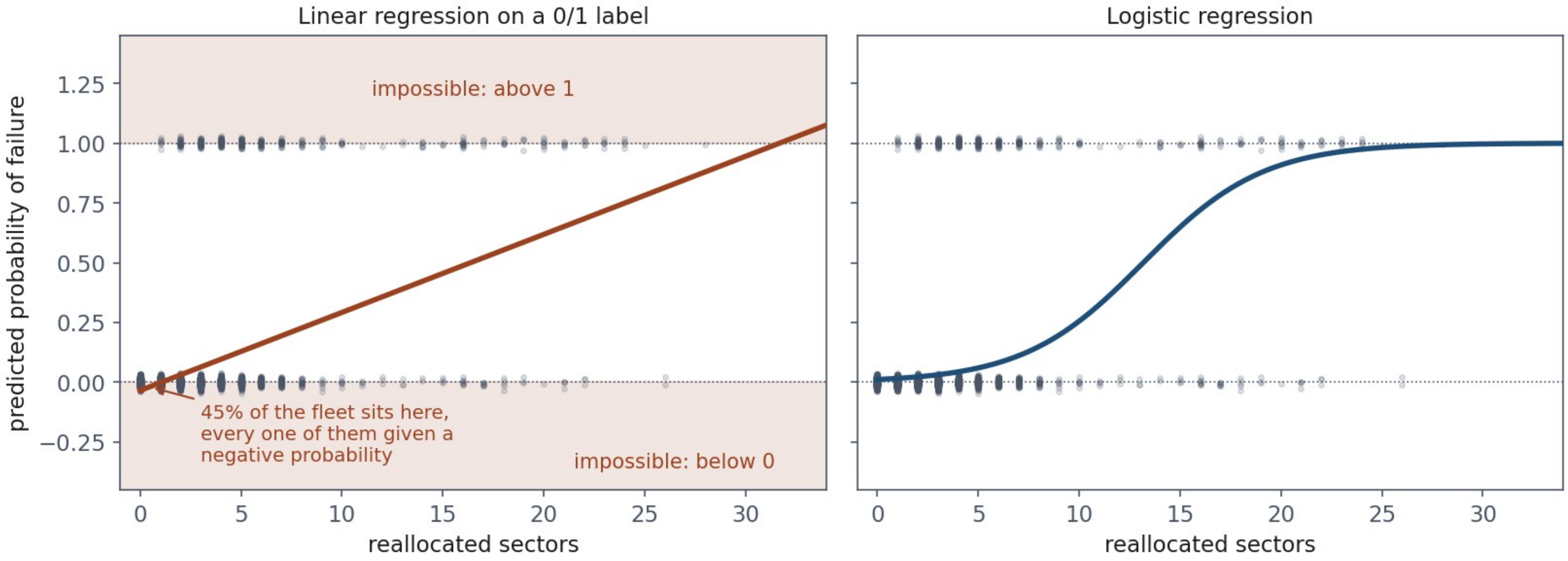
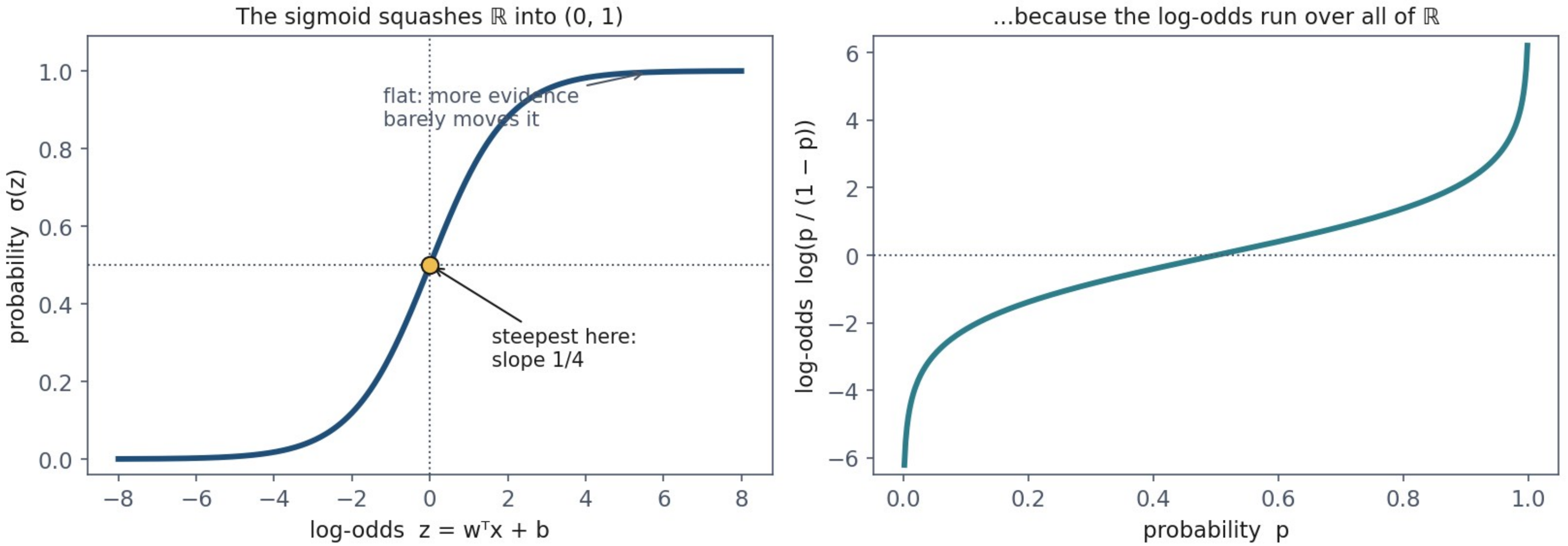
306

- **306 of the 8,000 drives failed.** 3.8% of the fleet
- Which means 96.2% did not

What goes wrong if we just fit a line



The sigmoid



The sigmoid squashes ℝ into (0, 1)
flat: more evidence
barely moves it
steepest here:
slope 1/4
probability σ(z)
log-odds z = wᵀx + b
1.0
0.8
0.6
0.4
0.2
0.0
−8
−6
−4
−2
0
2
4
6
8
...because the log-odds run over all of ℝ
log-odds log(p / (1 − p))
probability p
6
4
2
0
−2
−4
−6
0.0
0.2
0.4
0.6
0.8
1.0

The model, in one line

$$P(y = 1 \mid x) = \sigma(w^\top x + b), \qquad \sigma(z) = \frac{1}{1 + e^{-z}}$$

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What the coefficients mean

- The model is linear in the log-odds, so `exp(coefficient)` multiplies the **odds**
- Features standardised, so one unit = one standard deviation
- A coefficient of 1.80 means **six times the odds** of failing

The truth, and what we recovered

Feature	True	Fitted	Odds ×
reallocated_sectors	1.80	1.66	5.28
spin_retry_count	1.15	1.17	3.22
read_error_rate	0.85	0.87	2.39
temperature_c	0.45	0.47	1.59
power_on_hours	0.35	0.38	1.46
seek_error_rate	**0.00**	**0.02**	**1.03**

The intercept is the base rate

- Fitted intercept: **−6.09**
- σ(-6.09) = 0.0023 — a 0.2% chance for an average drive

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Fitting means minimising something — but what?

- Squared error worked for lesson 3
- Why not just apply it to the sigmoid’s output?

Maximum likelihood, in one sentence

- The model has a knob; each setting assigns a probability to what happened
- **Turn the knob so that what actually happened is least surprising**

Cross-entropy, also called log loss

$$J(w,b) = -\frac{1}{m}\sum_{i=1}^{m}\left[\, y^{(i)}\log p^{(i)} + \left(1 - y^{(i)}\right)\log\left(1 - p^{(i)}\right)\right]$$

The gradient is the one you already know

$$\frac{\partial J}{\partial w_j} = \frac{1}{m}\sum_{i=1}^{m}\left(\hat{y}^{(i)} - y^{(i)}\right)x_j^{(i)}$$

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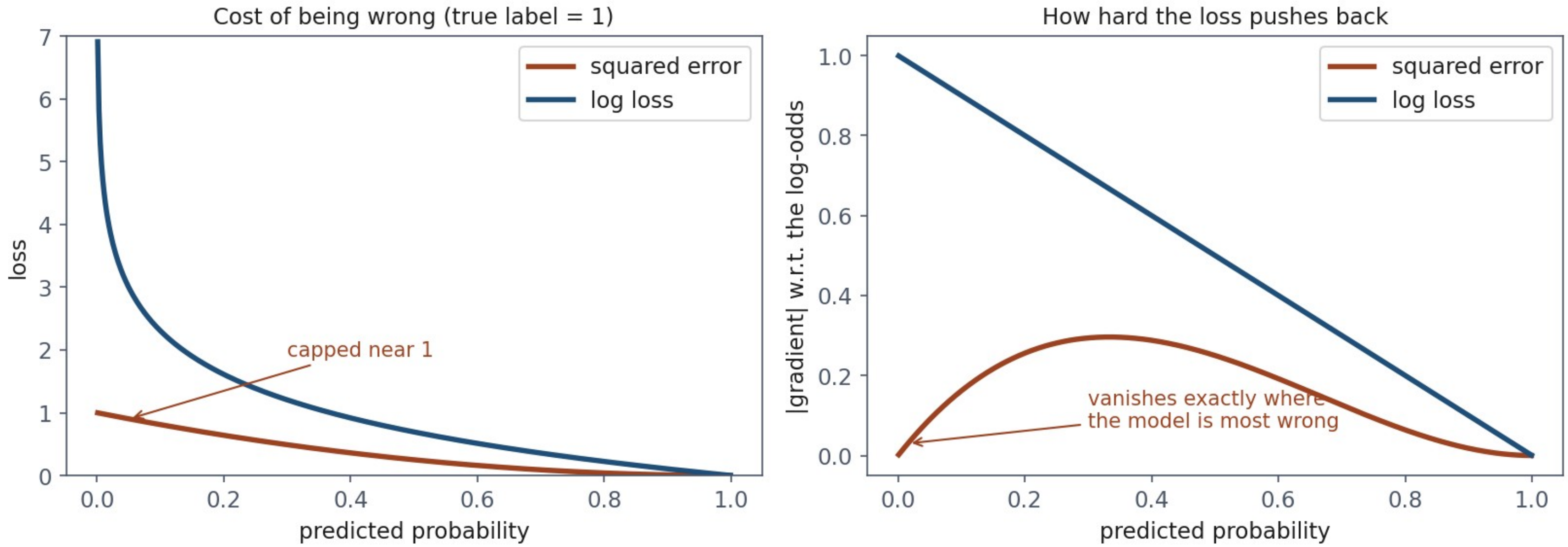
So why not squared error



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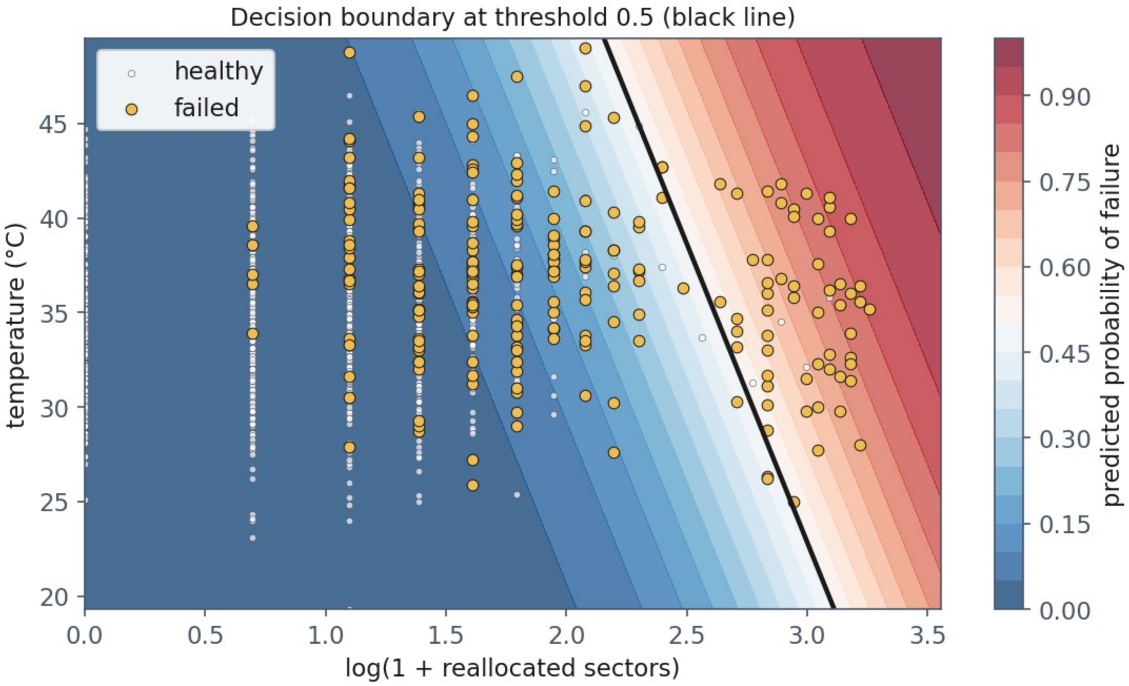
And it is convex, which squared error is not

- Log loss is **convex** in the weights, so any local minimum is global
- Squared error through a sigmoid is not

Notebook 1, live

- Fifteen lines of NumPy, then one line of scikit-learn
- Same answer

The decision boundary



Break

- Ten minutes

Two models

Model	Accuracy	Failures caught
Predict “healthy” for everything	**96.20%**	**0 of 76**
Logistic regression, threshold 0.5	97.70%	43 of 76

Why everyone makes this mistake once

- Accuracy is the fraction of predictions that were right
- On a **balanced** problem that is exactly what you want
- The interesting class is almost always the **rare** one

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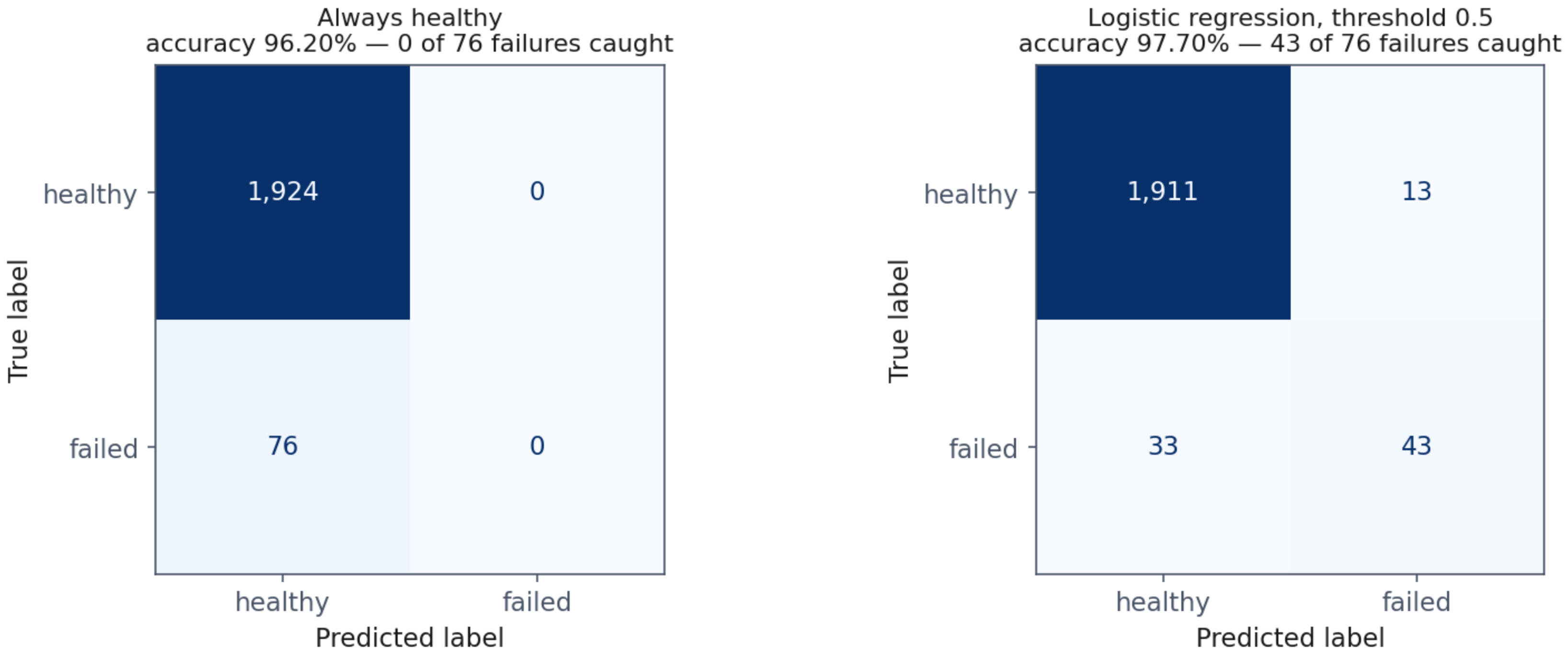
Four numbers instead of one



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The four outcomes, and what they cost

- **False positive** — we replace a healthy drive. A technician’s hour, €140
- **False negative** — a drive dies in service. Callout, rebuild, risk, €2,600

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Precision and recall: look at the denominator

Precision = TP / (TP + FP)
“of the alarms we raised”

	predicted healthy	predicted failing
actually healthy	TN 1911	**FP** 13
actually failed	FN 33	**TP** 43

Recall = TP / (TP + FN)
“of the drives that failed”

	predicted healthy	predicted failing
actually healthy	TN 1911	FP 13
actually failed	**FN** 33	**TP** 43

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Our model, in those terms

- Precision **0.768** — of 56 drives flagged, 43 really failed
- Recall **0.566** — of 76 drives that failed, we found 43
- Specificity 0.993 — of 1,924 healthy drives, we left 1,911 alone

F1: why the harmonic mean

Model	Precision	Recall	Arithmetic	Harmonic
Flag every drive	0.038	1.000	0.519	**0.073**

Read the minority class, not the average

- `weighted avg` F1: **0.975** — dominated by the healthy class
- `macro avg` F1: **0.820** — both classes get an equal say

Notebook 2, live

- The confusion matrix by hand, then from scikit-learn
- Then the threshold sweep

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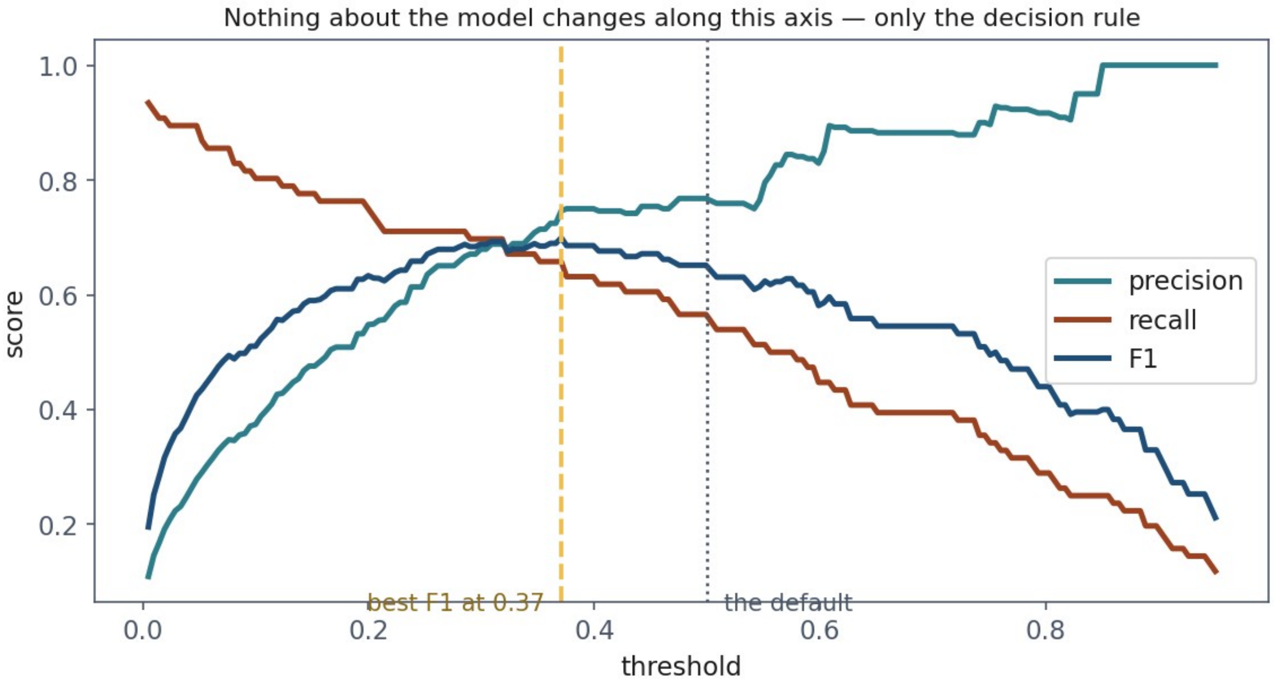
The threshold is a choice nobody made



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What the sweep actually costs

Threshold	TP	FP	FN	Precision	Recall	Accuracy
0.02	69	284	7	0.195	0.908	0.855
0.10	61	102	15	0.374	0.803	0.942
0.30	53	25	23	0.679	0.697	0.976
0.50	43	13	33	0.768	0.566	**0.977**
0.90	14	0	62	1.000	0.184	0.969

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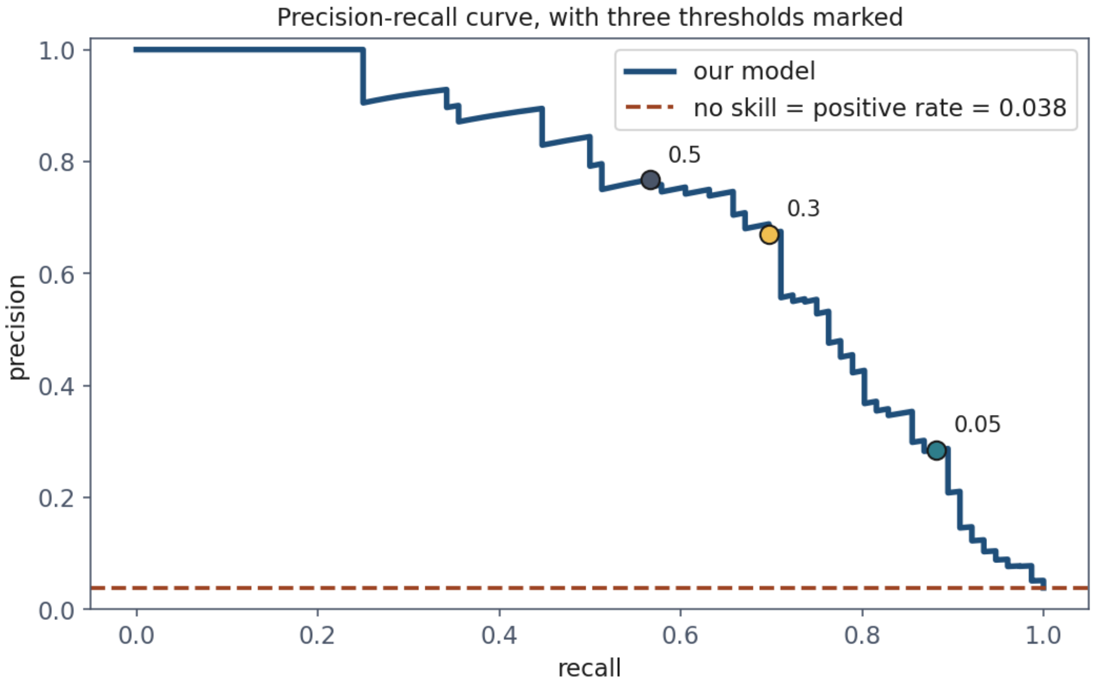
The precision-recall curve



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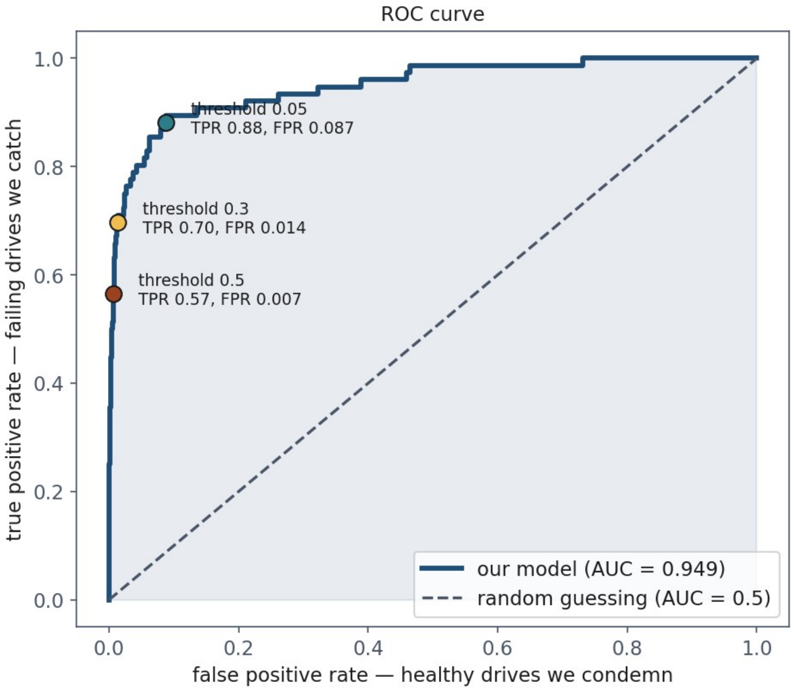
Comparing models without choosing a threshold

- We keep evaluating at one threshold — but which?
- What if we score the whole **ranking** instead?

The ROC curve



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What the area means

- **AUC = the probability that a random failing drive scores above a random healthy one**
- Ours: 0.949

What AUC does not tell you

- Divide every predicted probability by 10
- AUC is **unchanged**; every decision changes

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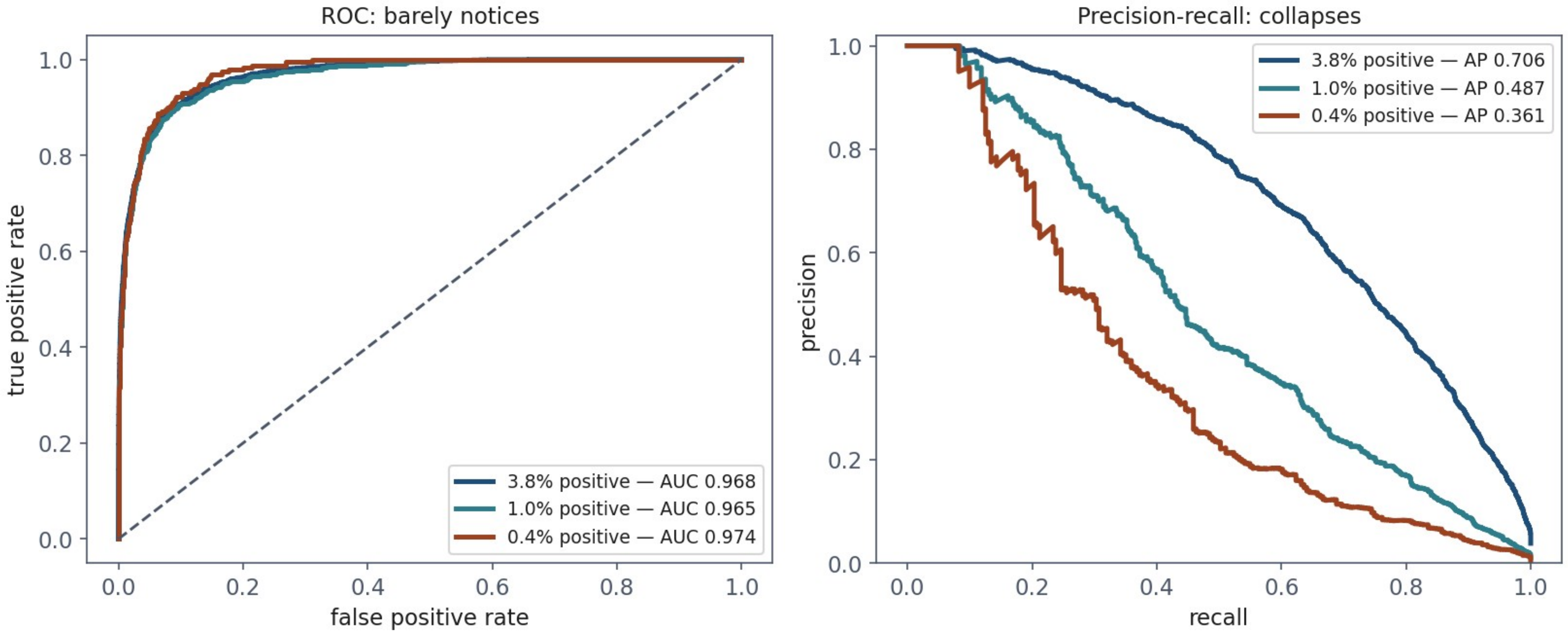
Where ROC misleads



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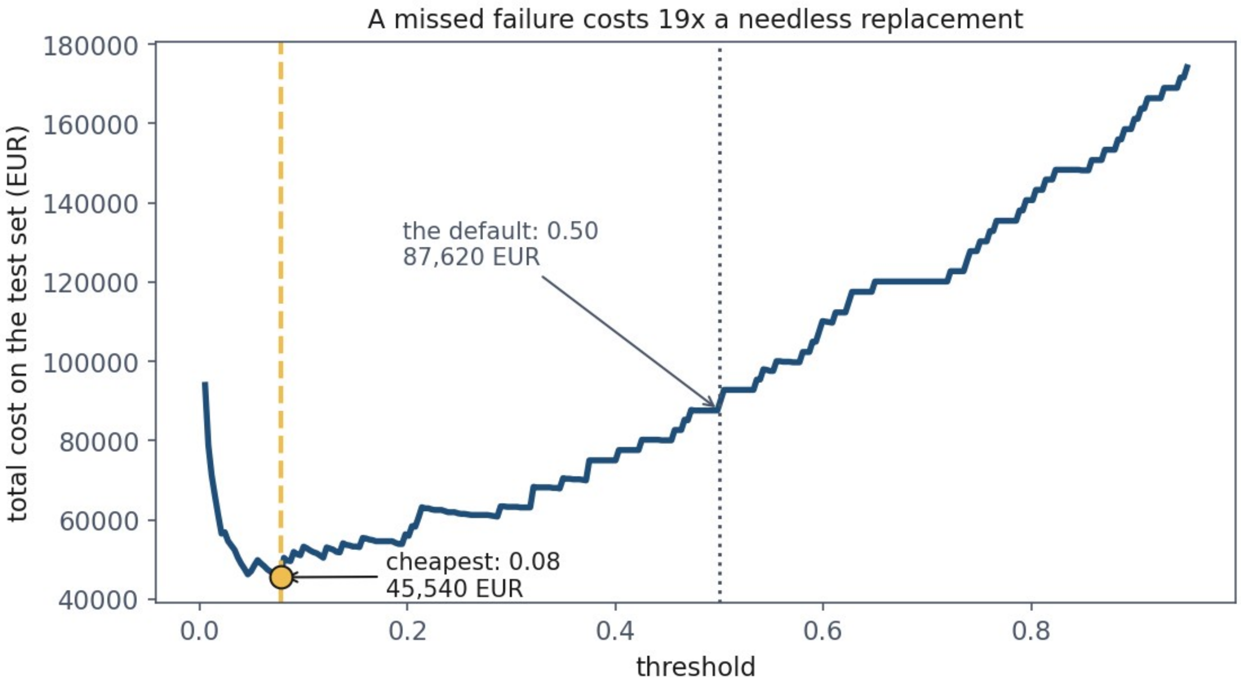
Why the two disagree

- FPR divides by **all healthy drives** — an enormous, growing denominator
- Precision divides by **the alarms raised** — where the false ones dominate

The rule

- On an imbalanced problem, report the **precision-recall curve alongside AUC**
- Never instead of it — and never AUC alone

Choosing the threshold from money



And it has a closed form

$$t^* = \frac{C_{FP}}{C_{FP} + C_{FN}} = \frac{140}{140 + 2600} = 0.051$$

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The caveat you must state

- We chose the threshold **by looking at the test set**
- That is precisely what lesson 1 forbade

Class weights: the same lever, pulled earlier

Model	Recall	Precision	AUC	Cost
Plain, threshold 0.50	0.566	0.768	0.949	87,620
Plain, threshold 0.08	0.855	0.349	0.949	45,540
Balanced weights, 0.50	0.895	0.249	**0.950**	49,500

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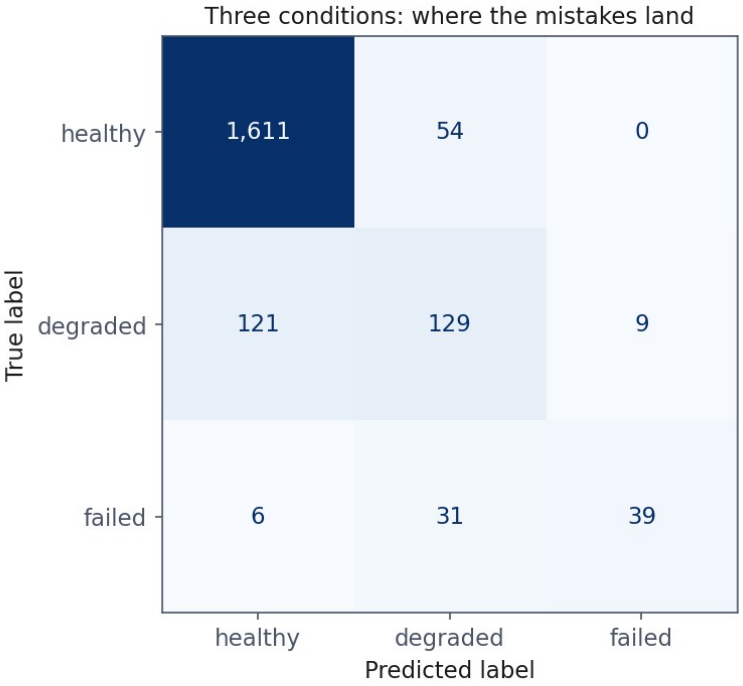
Notebook 3, live

- ROC by hand, the 200,000-pair experiment, the cost curve
- Then the multiclass matrix

More than two classes



Macro against weighted, one last time

- macro avg F1: **0.707**
- weighted avg F1: **0.883**

What to take away

- **96.20% accuracy, nothing caught.** Accuracy measures the majority class
- Precision is about the **alarms**, recall is about the **failures**
- The **threshold is a decision** — and it has a formula
- **AUC measures ranking** and hides imbalance

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Homework

- **Exercise 4**, due Friday 23 October
- Build a classifier, choose a threshold from a stated cost ratio, defend it